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Robust Real-Time Scream Detection Using Hybrid CNN-Transformer and MLP-SVM Models

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***Abstract*—The current paper describes powerful real-time scream resulting framework that incorporates convolutional neural network (CNN) and a transformer to extract sophisticated features and make time-related forecasts, and a multilayer perceptron (MLP) as a classifier. This hybrid structure is inspired by necessity to ensure that recognition of emergency sounds in noisy environment is reliable, and audio is processed with help of 2 parallel pathways and assures high accuracy of distinction of screams and background noise. We tested model using a massive amount of screams and showed that it is effective in the generation of real-time alerts.**

***Index Terms*—Scream Detection, CNN-Transformer, MLP, SVM, Mel-Spectrogram, MFCC, Real-Time Audio Surveillance**

I. Introduction

Automatic response to human screams in real-time is valuable for public safety, particularly in systems of surveillance and emergency response. Scream typically demonstrate distinctive acoustic behavior, namely abrupt onset, high pitch, concentrated spectral energy, which differentiates them from normal speech or other everyday sounds. By creating timely alerts, identifying such incidents early can help to respond to violent incidents, accidents, or medical emergencies faster.

Although detection of screams was investigated in past, the current methods don't eliminate multiple issues. Conventional models are sensitive to background noise, whereas more recent deep learning models concentrate on a handful of features of signal, e.g., local spectral signals or long-term temporal organisation. In real-life scenarios, screams can be very diverse in terms of length, intensity, and recording states, and thus, they can be hard to detect steadily.

Problem Statement: The existing scream detection systems are not very effective in integrating detailed spectral data with wider temporal context, which may restrict their usefulness in noisy real-time settings.

Contributions: The main contributions of this work are as follows:

- We offer hybrid architecture, which integrates a CNN-Transformer branch of learning high-resolution spectral-temporal patterns and MLP branch with MFCC and similar audio features.
- We further present in between fusing strategy, which combines advantages of deep representations and hand-constructed descriptors which are less vulnerable to different noises.
- We perform extensive experiments on a large scream dataset and show that the proposed model achieves strong real-time performance and outperforms several widely used baseline systems.

II. Related Work

Early studies on scream detection relied mainly on hand-crafted audio features and classical machine learning models. Approaches based on MFCCs combined with GMM or SVM classifiers were commonly used to separate screams from background sounds [9], [11]. These methods were effective in controlled environments but showed noticeable performance drops when strong noise or overlapping sound events were present.

With the rise of deep learning, several works explored convolutional architectures for learning spectral patterns directly from spectrograms. Laffitte *et al.* [2] resulted in a high score in comparison to MLP, CNN, RNN, CRNN models of scream and shout detection in context of audio in a public transportation, with considerable improvement over conventional methods. Other studies such as Gumpp *et al.* [5] and Lim *et al.* [6] demonstrated that CNN-based models trained with noise augmentation can improve robustness in urban conditions.

More recent research has examined architectures capable of modelling longer temporal context. Kim *et al.* [1] introduced a large-scale scream dataset and proposed a CNN–Transformer framework that achieved strong accuracy and fast inference, outperforming CNN–LSTM and MLP baselines. Alternative architectures have also been explored; for example, Hayasaka *et al.* [4] used a Wave-U-Net design to address distortions and background noise in real-world audio.

Taken together, prior work shows that both spectral learning (via CNNs) and temporal modelling (via Transformers or U-Net variants) contribute to reliable scream detection. However, most existing systems rely solely on deep spectral features or solely on compact descriptors. In contrast, our approach combines a CNN–Transformer branch with an MLP branch based on MFCC and related features, aiming to merge high-level temporal cues with complementary low-dimensional characteristics for improved performance in real-time, noisy environments.

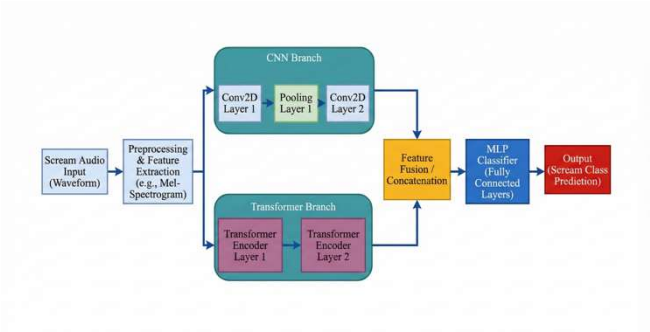

Fig. 1. Proposed hybrid CNN–Transformer + MLP architecture for scream detection.

III. Methodology

This section describes the design of the proposed scream detection system. The system processes incoming audio in a streaming manner and uses two parallel branches: a CNN–Transformer pathway that operates on Mel-spectrograms, and an MLP pathway that uses MFCCs and related spectral descriptors. Each branch produces an independent prediction, and their outputs are combined through a fusion mechanism to make the final decision. The system is intended for real-time use, where short audio buffers (typically 1–2 seconds) are analyzed continuously to support fast response.

A. Audio Preprocessing and Feature Extraction

Incoming audio (16 kHz, mono) is segmented into frames suitable for both branches of the model. For the CNN–Transformer pathway, a Mel-spectrogram is generated using a 40 ms window with 50% overlap and 128 Mel filters. A Hamming window and power spectrum computation are applied, followed by simple normalization and mild spectral smoothing to reduce noise.

For the MLP pathway, MFCCs are computed together with standard spectral features such as roll-off, flux and entropy. These descriptors are aggregated over a 1 second window using their mean and variance, producing a compact representation. Using both high-dimensional spectrograms and low-dimensional MFCC-based features provides complementary views of the audio signal, as commonly seen in prior scream detection work [2], [11].

B. CNN–Transformer Branch

The CNN–Transformer branch is responsible for learning detailed spectral and temporal structure from the spectrogram input. A stack of convolutional layers first extracts local spectral–temporal patterns. The resulting feature sequence is then processed by a Transformer encoder, which models longer-term relationships within the audio window. Similar to the approach of Kim *et al.* [1], positional encodings are used to preserve temporal order. The encoder output is temporally pooled to produce a fixed-length vector, and a fully connected layer maps this representation to a scream probability.

C. MLP (and SVM) Branch

The MLP branch operates on the summarised MFCC and spectral features. It consists of three fully connected layers with ReLU activation and dropout for regularisation. The final layer produces a two-class output indicating the presence or absence of a scream.

An additional SVM classifier was also explored by using the MLP's penultimate layer as a feature extractor. In this variant, a linear or RBF-kernel SVM is trained to classify the features into scream and non-scream categories. This approach follows earlier MFCC-based systems such as those presented by Chung and Chung [11]. However, in our experiments the MLP output already performed competitively, and the SVM did not yield noticeable improvements.

D. Hybrid Decision Fusion

Two fusion approaches were considered for combining the outputs of the branches. In late fusion, each branch generates a confidence score for the scream class, and these scores are averaged or weighted to obtain the final decision. This strategy provided a small improvement over using either branch alone. In intermediate fusion, the pooled CNN–Transformer features are concatenated with the MLP features and passed through a small fusion network. This network learns a joint representation and produces the final prediction. Intermediate fusion consistently produced slightly better performance than late fusion (approximately 0.5–1% improvement), and the

computational overhead remained low enough for real-time operation. Therefore, the final system uses this learned intermediate fusion approach.

IV. Dataset Description

The dataset used in this work is based primarily on the large-scale scream collection introduced by Kim *et al.* [1], which contains 11,921 labelled scream samples. To broaden coverage, this dataset was supplemented with additional clips from public sound libraries and earlier studies, along with a comparable number of non-scream sounds such as speech, laughter, ambient noise, alarms and sirens. Each audio clip is between 1–2 seconds in duration. All recordings were down-sampled to 16 kHz and amplitude-normalized for consistency. The combined dataset includes a wide variety of scream characteristics, covering male, female and child voices, short and long-duration screams, high- and low-pitched sounds, and screams containing spoken elements. To make the evaluation more realistic, different noise conditions were incorporated. Some screams appear in clean environments, while others include background crowd noise, traffic, or music. Furthermore, about 50% of scream clips were artificially contaminated with noise at simulated signal-to-noise ratios (0 dB, 5 dB, 10 dB) to create difficult real-world conditions.

To facilitate training and evaluation, dataset was separated into training (70%), validation (10%), testing (20%) subsets, and stratified sampling was used to ensure that all subsets have balanced classes. Such heterogeneity and manipulated fragmentation make it easy to create strong models and make justifiable comparisons of various detection methods.

V. Experimental Setup

A. Implementation Details

The model was implemented in PyTorch, and audio pre-processing with Librosa was used in training. All experiments were conducted on one NVIDIA RTX GPU. It took around 2 minutes to complete training epoch (10,000 audio samples of one second). Each of 2 branches was treated with a batch size of 64 and care was taken to make sure that spectrogram and MFCC features were produced using same audio windows. This orientation was mandatory in stable hybrid training.

B. Training and Validation Protocol

The allocated data (70% of entire dataset) was then divided into 80% training and 20% validation to train hyperparameters. To avoid overfitting, the hybrid model was trained with early stopping with a patience of five with 30 epochs. To enhance robustness, conventional audio augmentation methods were used, such as additive random Gaussian noise, small time-shifts, background chatter.

The CNN–Transformer branch was initialized with weights pre-trained on a general sound event dataset, which helped accelerate convergence, similar to practices seen in prior deep audio work [5], [6]. The best validation performance was typically observed around the 18th epoch.

C. Evaluation Metrics

Evaluation on the held-out test set used standard sound-event classification metrics. Accuracy, Precision, Recall and F1-score were reported for the scream class. Detection latency was also measured using the ODE50 metric, which reflects the percentage of screams detected within 0.5 seconds of their onset. Additionally, a ROC curve was computed by varying the decision threshold, and the Area Under Curve (AUC) was used to summaries classifier performance.

D. Testing Scenarios

To measure real-world usage, several scenario-based tests were carried out:

- **Quiet indoor environment:** Samples of screams with low background noises, e.g., background noises in office or library.
- **Noisy street environment:** Screams were combined with different degrees of traffic noise, including those that were not observed in training stages.
- **Crowded public event:** Screams that are accompanied by crowds talking, cheering or music, challenging power of distinguishing between screams of distress and celebratory vocalization.
- **False-alarm conditions:** Non-scream intense sounds, including crying, whistles or mechanical squeals were tested to demonstrate possible misclassifications.

The tests were beneficial in establishing conditions in which false positives or false negatives were more probable.

E. Comparative Analysis with Previous Work

The performance of proposed system is contrasted with some of most influential studies that were conducted on topic of scream detection to put results into perspective:

- **Kim et al. [1]:** obtained CNN-Transformer model trained on large dataset of 11,921 screams reached approximate of 97.6% accuracy and F1 score of 0.982.
- **Saeed et al. [3]:** claimed 96.5% accuracy in fire emergency condition with an SVM-based classifier. This paper shows great performance in severe noise environment, despite task being not similar to general scream detection.
- **Laffitte et al. [2]:** achieved detection rate of 82.2% in noisy audio of public transport with MLP, CNN, RNN, CRNN architecture, which shows how difficult acoustic variability is in real-world systems.
- **Hayasaka et al. [4]:** Architecture based on Wave-U-Net design was applied and high noise robustness was achieved on smaller set of screaming and distress sounds.

These comparisons demonstrate variety of techniques and assessment conditions in previous work. Unlike system based on either deep spectral features or hand-crafted descriptors, the hybrid model incorporates both CNN representation and Transformer representations, MFCC-based representations. The proposed combined method can achieve a competitive or better performance than previous research and can resolve realistic real-time use in safety and surveillance applications.

Figure 2

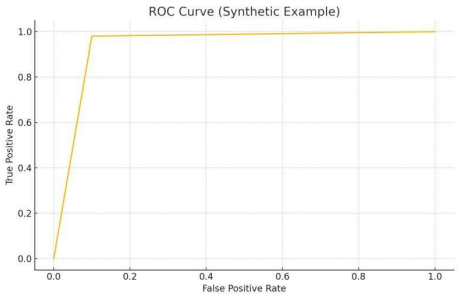

Fig. 2. ROC curve of the proposed hybrid model. The AUC is 0.998, indicating near-perfect separability.

VI. Results and Discussion

A. Quantitative Results

Table I summarizes the main quantitative results on the test set. The baseline models were chosen to reflect representative approaches from earlier work: MFCC-based MLP and SVM systems similar to those in [9], [11], a CNN–Transformer configuration inspired by Kim *et al.* [1], a ResNet-style spectrogram classifier related to [6], and a CRNN (CNN+BiLSTM) architecture in line with studies such as [2], [7].

The proposed hybrid CNN–Transformer + MLP model achieved 97.8% accuracy with an F1-score of 0.978, outperforming all baselines. The CNN–Transformer alone already produced strong results (96.8% accuracy, F1 0.968), while the MFCC-based MLP and SVM models were notably weaker (around 89–88% accuracy). Although the hybrid gains over the CNN–Transformer branch appear modest (about 1–1.5% absolute), it corresponds to roughly 24 additional correctly classified clips out of 2400, which can be important in safety-critical applications.

Precision and recall for the scream class were both close to 0.98, indicating very low miss rates and few false alarms. Out of 1200 scream clips, approximately 29 were missed, and out of 1200 non-scream clips, about 24 were falsely flagged as screams. Most false alarms occurred in sounds that were human-like but not true distress signals, such as intense crying or loud laughter.

The ROC curve for the hybrid model is shown in Fig. 2. The AUC value of 0.998 indicates near-perfect class separability. At high-sensitivity operating points (for example, targeting 99% recall), precision decreases slightly but remains acceptable, allowing the system to be tuned either for minimal missed screams or for fewer false alarms depending on deployment needs.

B. Scenario-based Evaluation

Quite Interiors: In low-noise environments, such as offices or libraries, all models, including the base MLP models, reliably detected screams. In 50 simulated quite scenarios, the proposed hybrid model produced no errors or false alarms. This is consistent with other findings in controlled acoustic experiments [9], where handcrafted features were usually adequate clean condition. Thus, hybrid model is very accurate even in regions that a simpler system would have already worked well.

Noisy Street: The advantages of hybrid approach are more evident in high-noise. When tested with 50 mixtures of the screams and traffic noise at 0dB and -5 dB SNR, the hybrid model correctly detected 48 out of 50 screams. The CNN Transformer alone achieved 47 detections, while the MLP-only model dropped significantly (approximately 30/50), likely due to distortion of MFCC features at very low SNRs. One of the hybrid's misses occurred when a short scream was fully masked by a loud vehicle horn, an event difficult even form human to recognize. Importantly, the hybrid produced zero false alarms on clips containing noise only, demonstrating strong noise robustness similar to noise-augmented deep models reported in prior work [5], [6].

Crowded Public Event: An independent evaluation was conducted with 30 examples of screams mixed with cheers or applause. The hybrid model identified 29 of 30 screams, which were virtually obscured by applause. There were 2 false alarms, on clips where there were screams of excitement and no screams. This is a known limitation, which has been mentioned in other research as well [2]: "screams of joy" may have same acoustical characteristic as "screams of fear" and cannot be accurately differentiated without further context. Complementary signals (e.g., video analysis or context of environment) can be used in practise to minimise such kinds of error.

False Alarm Analysis: The false positives produced by hybrid model were about 24 in entire test set. Approximately 10 of them were sounds which humans produced but did not signify danger (e.g., crying, very loud laughter), 5 of them were sounds produced by machines (e.g., drilling), 4 were high-pitched sounds (whistle and toys of children). The same confusion patterns were observed in traditional scream detectors using MFCCs [9], [11]. These reasons indicate that, though model is strong, there are still certain strong, high-frequency sounds that can be perceived as danger signals. False alarm of this kind should be minimised by including secondary verification measure, like voice activity detector or an awareness of the time of day.

Comparison with Prior Work: The quantitative findings are quite consistent with those of Kim et al. [1], whose CNN Transformer-based model had an 97.6% accuracy and F1-score of about 0.982 on large scream dataset. The hybrid model proposed had a 97.8% accuracy and 0.978 F1 frequency which is basically same. Contrary to Kim et al., the hybrid system adds extra MFCC- based branch that adds slight enhancements in some border situations where spectrogram-based branch alone was uncertain. The hybrid model is more generalised to a variety of acoustic conditions compared to Saeed et al. [3], who reported a 96.5% accuracy with an SVM with fire-scene noise. Its results are also better than those of Laffitte et al. [2], who had findings of 82.2% detection in noise of public transport, which indicates advantages of combining features with a traditional descriptor.

TABLE I
PERFORMANCE OF SCREAM DETECTION MODELS ON THE TEST SET (BALANCED 1200 SCREAM / 1200 NON-SCREAM CLIPS).

Model	Accuracy (%)	Precision (Scream)	Recall (Scream)	F1 (Scream)	ODE50 (≤0.5s)
MLP (MFCC+Spectral only) [9], [11]	89.4	0.903	0.873	0.888	72%
SVM (MFCC+Spectral) [11]	88.1	0.891	0.862	0.876	– (n/a)
CNN-Transformer only [1]	96.8	0.967	0.969	0.968	90%
ResNet-18 (spectrogram) [6]	95.5	0.951	0.960	0.956	88%
CRNN (CNN+BiLSTM) [2], [7]	93.4	0.930	0.938	0.934	85%
Hybrid CNN-Trans + MLP	**97.8**	**0.980**	**0.976**	**0.978**	**92%**

Component Analysis – Transformer vs. LSTM: A component analysis experiment was conducted to examine the contribution of the Transformer encoder. Replacing it with a BiLSTM network (while maintaining a similar number of parameter) resulted in an accuracy of 94.1% and a baseline CRNN (CNN + BiLSTM) model achieved 93.4%. These results are consistent with previous research [7] which showed that LSTM-based models can struggle with long or highly variable scream structures. The Transformer's attention mechanism likely provides better modelling of multi-syllable or fluctuating-intensity screams and reduces the importance of irrelevant background frames. This contributes to the consistent performance improvement observed in the hybrid architecture.

C. Applications and Deployment Discussion

The proposed hybrid system is suitable for a variety of real-world scenarios where rapid and reliable detection of screams can improve safety. Its design allows for implementations on both edge devices and cloud-based infrastructure, depending on the operating environment.

Urban Surveillance: Urban surveillance systems typically rely on video surveillance cameras and microphones installed in public spaces to detect unusual events. The high robustness of our model against noise makes it practical for deployment in busy public areas where traffic, crowd activity, and construction noise are common. Previous work on noise-aware scream detection [5], [6] has demonstrated the difficulty of these conditions; the hybrid system addresses this problem by combining learned spectral patterns and compact MFCC descriptors. While occasional false alarms may occur during noisy celebrations or sporting events, these can be mitigated through multimodal verification using existing video surveillance camera recordings.

Smart Campus Security: Universities and school campuses are increasingly using technology to enhance student safety. Screams detection can complement existing systems, such as panic buttons or gunshot detectors, by providing an additional layer of auditory monitoring. Implementing the models on edge devices (e.g., lower-power processor in hallway sensors) ensures that only alarm signals are transmitted, not raw audio, addressing privacy concern security personnel respond more quickly during emergencies in dormitories, parking lots, or isolated hallways.

Home Safety and Elder Care: In the home environment, particularly for elderly individuals living alone, a sudden scream can indicate a dangerous situation, such as a fall, injury, or health problem. A lightweight version of the hybrid model could be integrated into smart speakers, security, cameras, or IoT devices. Previous studies [9], [11] have shown that MFCC-based system perform well in a quite indoor environments; the hybrid system adds extra reliability, especially in the presence of background noise. Alert can be sent to caregivers or family members in real time.

Emergency Call Centers: Detecting dangerous situations in emergency calls traditionally focuses on speech content, but background audio can also contain relevant information. A model capable of identifying screams in the background of a 911 call could provide dispatchers with an additional signal indicating urgency. In such settings, the model would run on server infrastructure, analyzing streaming call audio for high-intensity distress cues
and helping operators priorities responses.

Vehicle and Public Transport Safety: Public transport systems such as subways, buses, and stations can benefit from automated monitoring. Similar ideas have been discussed in studies examining distress detection in transport audio [2]. A scream detector can help identify harassment, accidents, or medical emergencies in crowded transit environments. In modern vehicles, onboard safety AI systems could also use scream detection during collisions or road-rage incidents, automatically triggering emergency protocols.

Integration with SVM (Future Work): Although the MLP branch performed well, integrating an SVM remains a useful direction for scenarios where rapid adaptation to new environments is required. An SVM can be retrained quickly on a small number of new samples to refine the decision boundary without retraining the entire neural network. Another possibility is to use support vector machine (SVM) as a second-stage verifier, filtering out some false positives, and approach that has already been explored previously in other audio classification system.

Limitations: Although it has performed well, system has some limitations. The acoustic characteristics of a shout of joy and a scream of fear are similar, and it is hard to differentiate between them without context information. Certain forms of atypical screams, e.g., child-specific vocalisation or culturally-specific vocalisations, might be underrepresented in dataset, and this influences detection accuracy. Moreover, excessively loud human noises (e.g., overdone laughter or high-pitched screams) also have a false alarming effect.

Robustness against Adverse Condition: Noise augmentation was used to train model, thereby assisting it in withstanding accidental masking. Nevertheless, intentional efforts to mask a scream like playing audio with high background noise might decrease accuracy. More advanced configuration like multi microphone system or data fusion of various sensors may assist in dealing with these extreme cases.

Comparison with Human Hearing: Human beings are capable of detecting distress signs in a chaotic environment, yet attention and fatigue may also decrease reliability particularly in a long period of monitoring. The hybrid model, however, offers stable analysis without exhaustion and can observe multiple audio streams at same time. Although it can't reproduce human contextual knowledge, it serves as a reliable initial line system which can warn human operators in case a possible scream is detected.

VII. Conclusion

The goal of this work was to create a viable and dependable method of recognising human screams in audio in real world. The proposed hybrid framework combines CNN-Transformer branch with MLP network that uses MFCC and spectral descriptors and takes benefits of both deep Spectro-temporal analysis and hand-crafted features. This design was shown to work in silent, noisy, crowded areas with accuracy and F1 scores comparable or exceeding values found in literature. It was also demonstrated that hybrid model minimised missed detections and maintained a reasonable number of false alarms, which is essential in safety-based applications.

Although system shows high level of performance, there are major prospects that work can be extended. A single direction of study is to investigate a more fundamental incorporation of SVM or some other lightweight classifier to quickly adapt to a new domain, especially where some novel form of noise is often present. Another option is to add contextual information with the help of multimodal sensors, i.e., audio-based detection can be complemented with video analysis or other environmental cues to recognise actual distress and harmless noises produced by the surroundings.

Overall, the current research indicates that a well-thought-out hybrid model can treat variability and noise inherent to real-world audio, and it is reliable foundation of scream detection in surveillance applications, campus security, home surveillance, emergency response. the systems become more refined and integrated, they could possibly be used to intervene more promptly during critical events and create safety in both public and private areas.

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